

SOS System: One-Tap Tourist Emergency Routing with Haversine Nearest-Station Assignment

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SOS System -- Tour Ceylon -- Unified Tourism Platform for Sri Lanka

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Abstract

Tourists in an unfamiliar country rarely know which police desk covers their GPS coordinate. This paper describes Tour Ceylon SOS: a progressive web app (port 5175) and a station-scoped police live desk (port 5176) on the same Flask API (port 5002). A single tap among six incident types requests a fresh high-accuracy geolocation (maximumAge 0, timeout 20 000 ms, no watchPosition) and creates an incident assigned to exactly one dispatchable station. Assignment is nearest neighbour on the WGS84 sphere with Earth radius 6371 km (Haversine). Of 73 police rows loaded from Excel, coordinate clustering at 4 decimal degrees (~11 m) yields 44 dispatchable desks; duplicate headquarters and rows without coordinates are excluded. A hospital registry of 42 facilities (40 with coordinates, 2 hotline-only) supports a separate dialer; hospitals are not auto-dispatched. Workflow is strictly forward: open, acknowledged, dispatched, closed. Chat, encrypted PII (Fernet), optional Twilio/SendGrid, and a closed-case PDF complete the loop. No machine-learning accuracy is claimed. Operational metrics (volume by day, minutes-to-acknowledge on closed cases) are exposed to officers. Limitations include great-circle versus road distance and one-shot location (continuous ping returns HTTP 409).

Index Terms-- emergency response, Haversine, nearest neighbour, geolocation, PWA, incident workflow, Sri Lanka Police, Leaflet

I. INTRODUCTION

Emergency numbers and local police geography are not obvious to visitors. Tour Ceylon SOS converts a typed emergency and a GPS fix into a case at the nearest dispatchable desk, then lets that desk acknowledge, dispatch, and close the case while chatting with the tourist. The design goals encoded in source comments are: (i) do not spam all 73 loaded desks; (ii) capture location only at SOS time; (iii) encrypt passport/phone at rest; (iv) reuse Tour Ceylon login so tourists do not register twice.

II. SYSTEM CONTEXT

The tourist client is React 18 + Vite 5 with vite-plugin-pwa 0.20.5 (Workbox NetworkFirst for API). The police client adds Leaflet 1.9.4, react-leaflet 4.2.1, and Recharts 2.12.7. Both proxy /api to Flask. SOS tables are created by SQLAlchemy in seed_sos_data.py (they are not in the travel schema.sql). PII uses cryptography 42.0.8 Fernet. Maps use OpenStreetMap tiles. Icons: lucide-react 0.454.0.

TABLE I

SOS data loaded from backend/data/sos/ Excel workbooks after seed rules.

Entity	Count	Notes
Police rows (Excel)	73	One officer login per row
Dispatchable stations	44	After 4-d.p. coordinate clustering
Non-dispatchable desks	29	Shared HQ coords or missing lat/lng
Government hospitals	37	All geocoded
Private / hotline rows	5	3 geo + 2 hotline-only
Hospitals used for geo match	40	is_hotline_only = false
Incident types	6	general, medical, rape, theft, accident, harassment
Workflow states	4	open -> acknowledged -> dispatched -> closed

III. METHODOLOGY

A. Client Geolocation

Hook `useGeolocation.getCurrent()` calls `navigator.geolocation.getCurrentPosition` with `enableHighAccuracy` true, `maximumAge` 0 (no cached fix), and `timeout` 20 000 ms. Horizontal accuracy is stored on `location_pings.accuracy` but is not a routing weight. If permission is denied, SOS aborts; there is no last-known fallback at trigger time (`last_known_lat/lng` on the tourist row is updated after a successful create).

B. Haversine Assignment

Let ϕ_1 , λ_1 be WGS84 latitude and longitude in radians. With $R = 6371$ km,

$$a = \sin^2(d\phi/2) + \cos(\phi_1) \cos(\phi_2) \sin^2(d\lambda/2), d = 2 R \arcsin(\sqrt{a}) \quad (1)$$

Function `nearest_entity` scans all candidates with non-null coordinates and returns the minimum d . Distance is stored to three decimal places as `distance_to_station_km`. There is no OSRM or Google Directions hop; terrain and one-way roads are ignored. If no dispatchable station remains, the code retries all stations that have coordinates.

C. Dispatchable-Desk Deduplication

`COORD_PRECISION = 4` rounds latitude and longitude (~ 11 m). Rows that share a rounded pair form a cluster; all but one are marked `is_dispatchable = false`. The survivor prefers a row with `Division` set, else the lowest id. Rows missing coordinates are never dispatchable. This prevents several Colombo HQ labels at the same pin from all receiving the same SOS.

D. Incident Types and Workflow

Types: general, medical, rape (UI label Assault; aliases rape/assault, assault), theft, accident, harassment. Unknown strings collapse to general. Status transitions are monotonic: a closed case cannot reopen; backward steps are rejected in `_transition`. Closing sets `tracking_active` false (already false by product policy). Police endpoints: PATCH acknowledge, dispatch, close. Tourist poll: GET status every 5 s. Police queue poll: 7 s with an audio cue on new open cases. Chat poll: 4 s. Police incident-detail poll: 8 s. Messages are scoped to the assigned `station_id` in the officer JWT.

E. Hospital Dialer versus Dispatch

GET `/api/hospitals/nearest` uses the same Haversine scan over geo hospitals. The tourist UI opens a tel: link. `incident.hospital_id` remains null; `hospital_notified_at` is never set in current routes. Hotline-only rows are a fallback if no geo hospital exists.

F. Privacy, Notifications, Export

Phone, passport/NIC, and emergency-contact fields are Fernet-encrypted. Continuous tracking was removed: POST `.../ping` returns 409. Notifications default to console logs (`NOTIFY_MODE`); Twilio SMS and SendGrid email are optional. Closed cases export via ReportLab (GET `.../report.pdf`). Travel-to-SOS identity uses POST `/api/tourists/bridge-travel`.

G. Approaches Considered

Road-network routing would better match how a patrol car drives; it was not integrated. Broadcasting every desk would maximise recall of someone seeing the SOS but would violate the spam constraint and the station-scoped JWT model. An ML classifier of incident type from free text is unnecessary because the tourist taps a labelled button. The selected stack is deterministic geometry plus a finite-state workflow--appropriate where false dispatch to the wrong town is more serious than a few extra kilometres of great-circle error.

IV. RESULTS

Routing is exact nearest neighbour among the 44 dispatchable coordinates; there is no confusion matrix against a jurisdictional ground-truth layer (none is in the repository). Table II lists measurable system parameters. GET `/api/incidents/stats` (officer JWT) returns `volume_by_day`, `minutes_to_ack = (acknowledged_at - triggered_at)/60` for closed cases with both timestamps, counts by type, and active/closed totals. Demo seed data uses synthetic acknowledge delays of approximately 5 min (medical) and 10 min (theft); those are fixtures, not a field trial. A micro-benchmark of Haversine nearest-neighbour over 44 synthetic coordinates (1000 repeats, CPython on the development PC) took 44.4 ms total, i.e. 0.044 ms per full scan. No automated unit test of Haversine was found in-repo.

TABLE II
Implemented SOS parameters and polling intervals.

Parameter	Value
Earth radius R	6371 km
Coordinate cluster precision	4 decimal degrees (~11 m)
Geolocation timeout	20 000 ms
maximumAge / watchPosition	0 / not used
Tourist status poll	5 s
Chat poll	4 s
Police queue poll	7 s
Police incident-detail poll	8 s
Distance storage	round(d, 3) km
Default notify channel	console (Twilio/SendGrid optional)
PWA plugin	vite-plugin-pwa 0.20.5
Map library	Leaflet 1.9.4
Haversine nearest-of-44 (bench, 1000 scans)	mean 0.044 ms / scan

End-to-end path: tap type -> overlay + optional vibrate -> getCurrent -> POST /api/incidents {tourist_id, type, lat, lng, accuracy} -> assign station -> one location_ping row -> notify station / emergency contact / hotel if present -> tourist workflow + chat -> police PATCH chain -> optional PDF.

V. DISCUSSION

A. System-Level Value

SOS is the safety backstop for the itinerary product: the same tourist record that booked moods and hotels can raise a typed emergency without explaining Sri Lankan police geography. Station scoping makes a live demo credible (one desk's queue, not a national flood).

B. Limitations

- Great-circle distance can rank a station across a mountain or lagoon ahead of a farther-but-faster road desk.
- GPS accuracy is stored but unused; indoor/urban-canyon error is unmodelled.
- 29 of 73 rows never receive SOS traffic by construction.
- One-shot location: the police map is not a live track (product decision, HTTP 409 on ping).
- Hospital is tourist-initiated dialling only.
- Polling (4-8 s) is not WebSocket push; officers may see new cases a few seconds late.
- Default notifications are logs unless SMS/email secrets are configured.
- No published field accuracy study versus actual Sri Lanka Police boundaries.

C. Future Work

OSRM driving time; polygon jurisdiction; availability/load of officers; WebSockets; offline PWA outbox; auto-notify nearest hospital on medical type; unit tests with known lat/lng fixtures and expected station_id.

VI. CONCLUSION

Tour Ceylon SOS assigns one of six typed emergencies to 1 of 44 dispatchable stations using Haversine nearest neighbour (R = 6371 km) on a one-shot high-accuracy GPS fix. It should be reported as a geospatial routing and workflow system. Classifier F1 is not applicable. The component's importance is a privacy-aware, station-scoped emergency path bound to the same tourist identity as the itinerary planner.

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